Esercizio 1. *Let $|L| < \kappa$. Then there exists a κ^+-saturated model M of $\mathcal{T}$ with $|M| < 2^\kappa$.*

Proof. We will construct a chain $(M_\alpha : \alpha < \kappa^+)$, defined as follows:

- Take M_0 to be any e.c. model of $\mathcal{T}$ with $|M_0| \leq \kappa$
- Take the union if limit
- To pass from M_α to $M_{\alpha+1}$ we need an auxiliary chain $(M_{\alpha,j} : j < 2^\kappa)$ and a dove-tail enumeration $e : 2^\kappa \to 2^\kappa \times 2^\kappa$:
  - Take $M_{\alpha,0} = M_\alpha$
  - Take the union if limit
  - To pass from $M_{\alpha,j}$ to $M_{\alpha,j+1}$ first enumerate the diagrams of the form $f : A \to B, i : A \to M_{\alpha,j}$ with A e.c. model of $\mathcal{T}$ and $B \models \mathcal{T}, |B| \leq \kappa$. If $e(j) = (r, s)$ then consider the r-th diagram enumerated for $M_{\alpha,s}$. Let such diagram be $f : A \to B, i : A \to M_{\alpha,s}$. Now consider $l : B \to B'$ be the direct limit construction. Observe that modifying the construction using downward l.s. at each step one can require $|B'| \leq |B|$. Now use amalgametion to get N e.c. model of $\mathcal{T}$ and use downward l.s. to get $M \prec N$ with $M_{\alpha,s}, B \subseteq M$ and $|M| \leq 2^\kappa$. Use the direct limit construction to arrive to $M_{\alpha,s+1}$ (as observed above we can bound the cardinality with 2^κ.

One can verify inductively that $\forall \alpha < \kappa^+ \forall j < 2^\kappa |M_{\alpha,j}| < 2^\kappa$. In particular this gives that $|M_{k^+}| \leq 2^\kappa$. Since κ^+ regular we have saturation (every diagram to be checked must have been enumerated somewhere).

□